

ECP257 ELECTRONIC CIRCUITS

MiniProject Report: Product Counting Machine

BT23ECE007,BT23ECE011,BT23ECE013,BT23ECE014

Introduction

The Product Counting Machine is an innovative device designed to automate the process of counting objects on a conveyor belt or production line. It leverages the capabilities of infrared (IR) sensors, transistors, and integrated circuits to accurately detect and count objects as they pass through a designated area. This automation solution offers several advantages, including:

- **Increased Efficiency:** Automates the counting process, reducing the need for manual labor and minimizing human error.
- **Improved Accuracy:** Ensures precise counting, leading to better inventory management and quality control.
- **Enhanced Productivity:** Streamlines operations and boosts overall productivity.
- **Cost-Effective:** Reduces labor costs and minimizes product loss.

Components Used and Specifications

1. **IR Sensor:**
 - Detects the presence or absence of an object by emitting and receiving infrared light.
 - Provides a digital output signal (high or low) based on the object's presence.
2. **10k Ohm Resistor:**
 - Limits the current flow in the circuit, protecting components and ensuring stable operation.
3. **BC547 NPN Transistor:**
 - Acts as a switch, amplifying the weak signal from the IR sensor to drive the IC4026 counter.
4. **IC4026:**
 - A decade counter that counts the number of pulses received from the transistor.
 - Outputs the count in binary-coded decimal (BCD) format.
5. **9V Battery:**
 - Provides the power supply for the entire circuit.
6. **7-Segment Display (Common Cathode):**
 - Displays the count in a readable format.
7. **Breadboard:**
 - A platform for prototyping and testing the circuit.

Working Principle

1. **Object Detection:**
 - The IR sensor emits infrared light towards the conveyor belt.

- When an object passes through the sensor's field of view, it interrupts the light beam.
- The sensor detects this interruption and generates a digital pulse.

2. Signal Amplification:

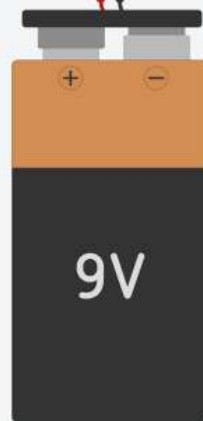
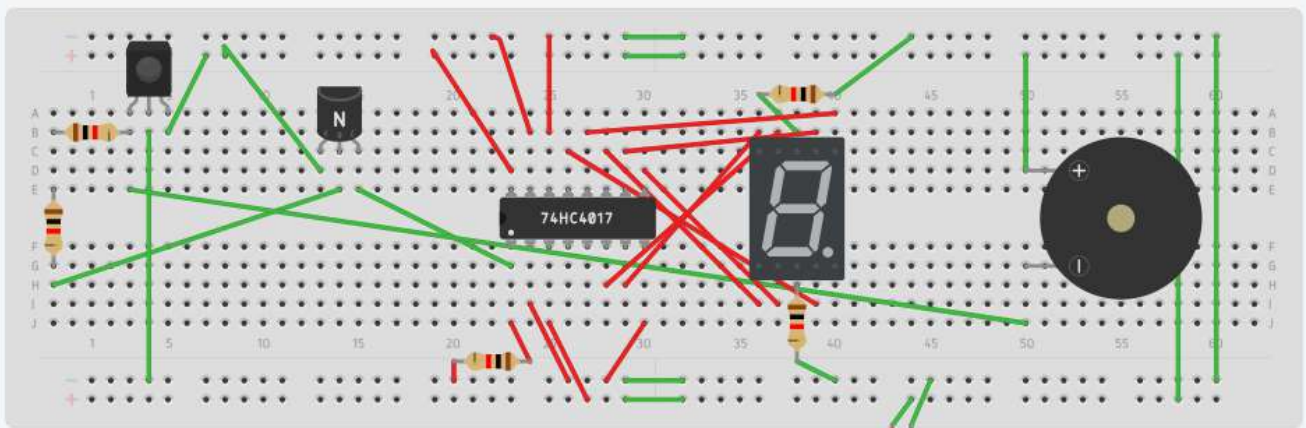
- The weak pulse from the IR sensor is amplified by the BC547 transistor.
- The amplified signal is then fed to the IC4026 counter.

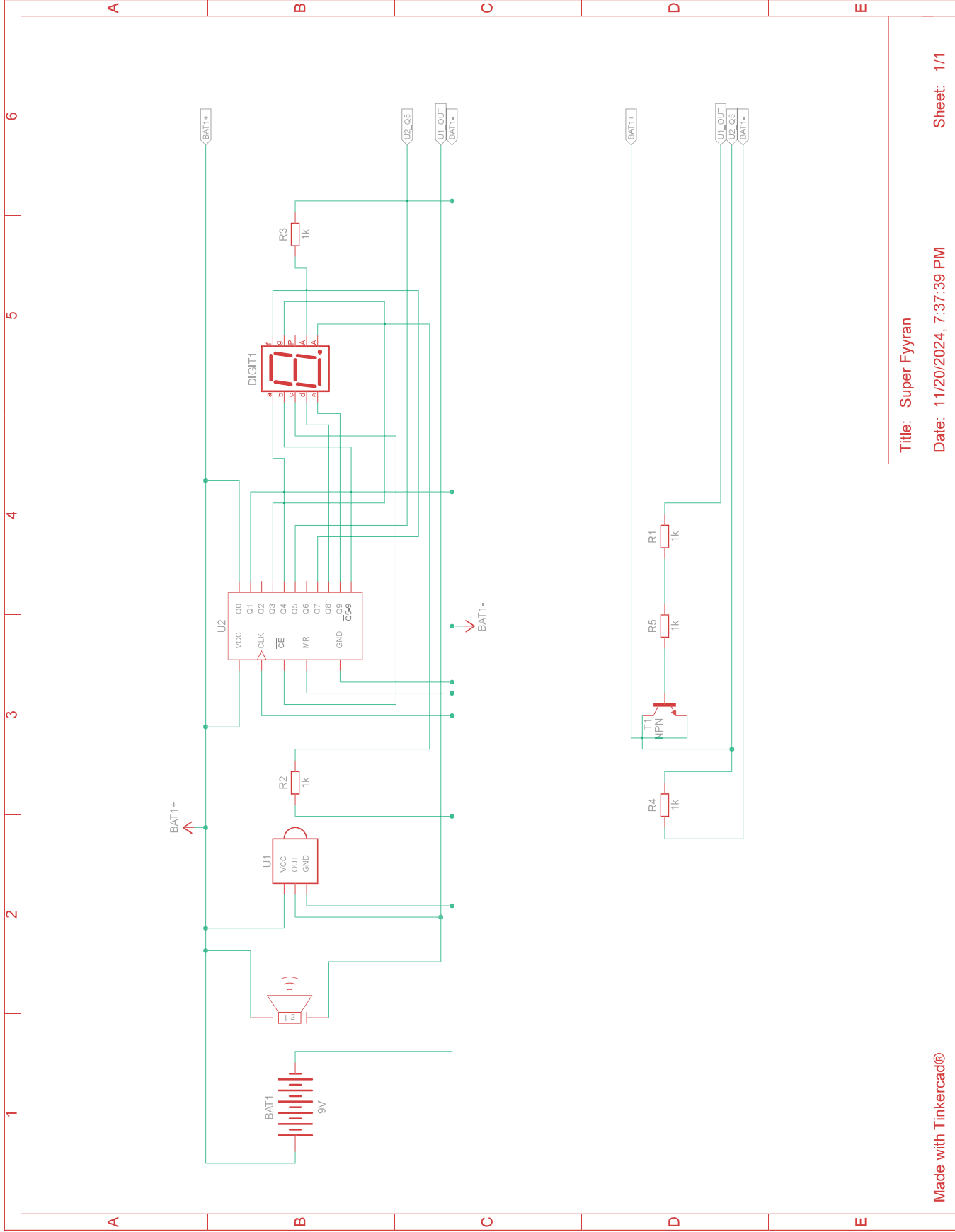
3. Counting and Display:

- The IC4026 counts the number of pulses received from the transistor.
- The count is displayed on the 7-segment display in decimal format.

Conclusion

The Product Counting Machine is a versatile and efficient solution for automating object counting in various industrial and commercial applications. By utilizing a combination of electronic components, this project demonstrates the practical application of fundamental electronics principles. This project can be further customized and enhanced to meet specific requirements, making it a valuable tool for improving productivity and accuracy in various industries.



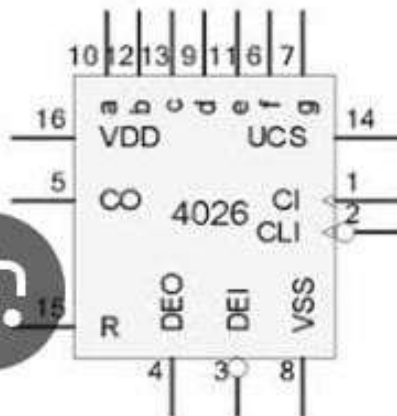


IC CD4026

CD4026 adalah IC yang berfungsi sebagai BCD counter sekaligus seven segment decoder



Simbol:



CD4026

